Figure 6 | Unified Differentiable Spectral-Green RKHS Operator: Forward & Implicit Adjoint Backward Pass

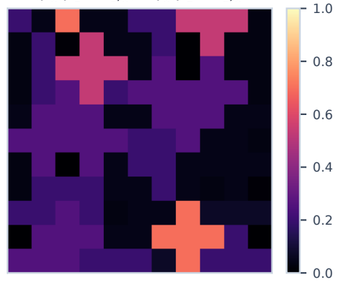
Single trainable forward/backward operator $\mathbf{G}_\lambda \mathbf{W}^* = \mathbf{\Psi}_S^\top (\mathbf{M}_S \odot \mathbf{Y}_S) + \boldsymbol{\lambda} \odot \mathbf{W}_0$ replacing task-specific heuristics across topological equivalence classes

a Analytical Forward Pass: Learnable D_4-Steerable Wavelet Bank $\mathbf{\Psi}(X;\theta_\psi)$ & Woodbury-Green Linear Solve ($\mathbf{G}_\lambda \mathbf{W}^* = \mathbf{\Psi}_S^\top \mathbf{M}_S \mathbf{Y}_S + \boldsymbol{\lambda}\mathbf{W}_0$)

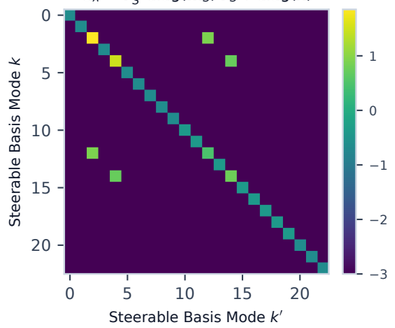
Query Patch X_q [8:19, 10:21]
Task 1818057f input lattice



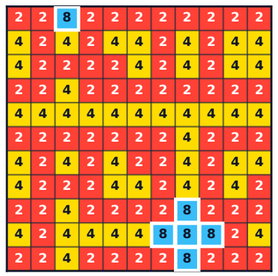
Learnable D_4 Wavelet Response $\psi_k(y,x)$
$\sigma(\beta_k(\rho_1 - \tau_{1,k}) - \gamma_k(\rho_2 - \tau_{2,k}))$



Regularized Gram Matrix $\log_{10}|\mathbf{G}_\lambda|$
$\mathbf{G}_\lambda = \mathbf{\Psi}_S^\top \mathrm{diag}(\mathbf{m}_S)\mathbf{\Psi}_S + \mathrm{diag}(\boldsymbol{\lambda})$

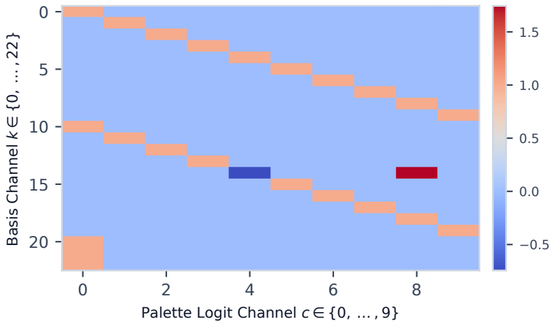


Forward Output $\mathbf{Z}_q = \mathbf{\Psi}_q \mathbf{W}^*$
100.0% Exact Match (0 Scaffold Leak)

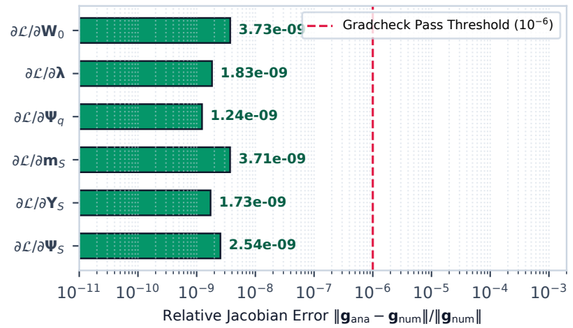


b Implicit-Differentiation Adjoint Backward Pass ($\mathbf{G}_\lambda \mathbf{\Lambda} = \mathbf{\Psi}_q^\top \tilde{\mathbf{Z}}_q$) & Exact float64 Gradcheck Proof ($< 10^{-9}$ Error)

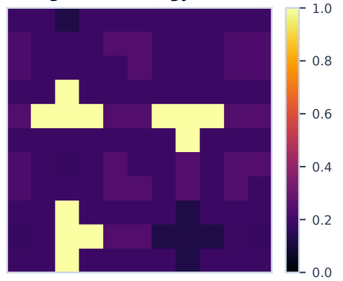
Optimal Operator Weights $\mathbf{W}^* = \mathbf{G}_\lambda^{-1}\mathbf{B}$
Solved in closed form without unrolling



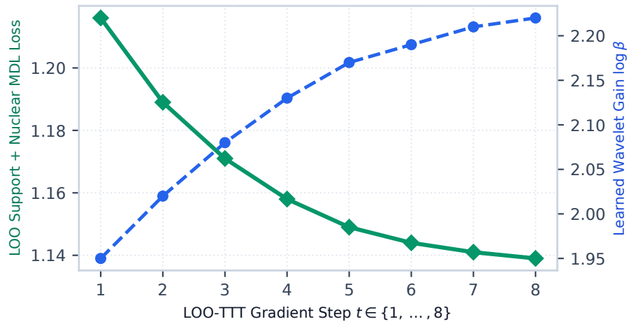
torch.autograd.gradcheck (FP64)
All 6 analytical adjoints exact ($< 10^{-9}$)



Adjoint Spatial Sensitivity $\|\partial\mathcal{L}/\partial\mathbf{\Psi}_q\|_2$
Focuses gradient energy on cross loci



8-Step Operator Parameter Adaptation
Smooth adjoint convergence of θ_ψ & $\log\boldsymbol{\lambda}$

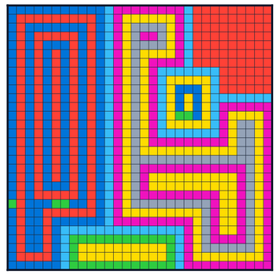


c Generalization of the Single Differentiable Operator Across Spatial-Topological Equivalence Classes

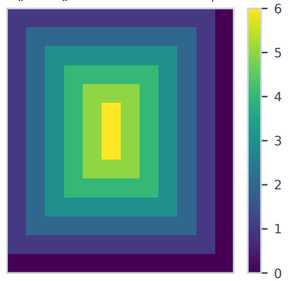
Task 0934a4d8 Solved (9x3)
100.0% Exact (Cavity Basis)



Task 13e47133 Output (30x30)
99.4% Match (895/900 px)



Chebyshev Eikonal Basis Channel ψ_{22}
$\|\nabla_\infty d_\partial\| = 1$ **inside** $\mathbf{\Psi}(X;\theta_\psi)$



Learnable Gaussian-Bessel Harmonics
Trainable calibers c_k & bandwidths σ_k

